# GABRIELLA ESEREOMO AVBAYERU

**GIS & Remote Sensing · Geospatial Analysis**

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Portfolio: mygabbbie.github.io/GIS-Portfolio · GitHub: github.com/MyGabbbie

## SUMMARY

Master’s student in Geoinformation Science with hands-on experience in satellite-based gas flare monitoring, environmental risk mapping, land cover classification, and spectral index analysis using Google Earth Engine and ArcGIS Pro. Independent project work featured by GIZ Nigeria. Seeking GIS, geospatial analysis, or remote sensing roles.

## EDUCATION

**MSc, Geoinformation Science - University of Ibadan** *Expected 2027*

**BSc, Urban and Regional Planning - Federal University of Technology, Akure** *2024*

## SKILLS

**GIS Software:** ArcGIS Pro, QGIS, Google Earth Engine

**Remote Sensing & Analysis:** VIIRS nighttime lights, land cover classification, change detection, Markov transition modelling, landscape metrics, spectral indices (NDVI, SAVI, NDWI, NDBI), multi-criteria risk modelling, hydrological & watershed delineation, spatial overlay & geoprocessing

**Data & Programming:** R (basic, coursework-level), Excel / tabular analysis, vector & raster data management

**Communication:** Technical & research writing, data storytelling, cartographic production, independent research design

**Languages:** English (Fluent)

## PROJECTS

**Nigeria After Dark - Nighttime Lights & Gas Flaring from Space** *2026*

*Independent project - Google Earth Engine, ArcGIS Pro*

- Built a VIIRS nighttime lights map of Nigeria separating city lights from Niger Delta gas flares; validated against World Bank / GFMR catalogued sites. Featured by GIZ Nigeria at a national gas flaring report launch. 21,000+ LinkedIn impressions.
- Follow-up analysis showed flared volume fell 29% since 2012 while intensity per barrel rose 9% - proving the decline tracked oil production, not improved gas capture.

**Cholera Environmental Risk Index - Nigeria** *2026*

*Independent project - Google Earth Engine, ArcGIS Pro*

- Built a composite environmental risk index from four normalized variables (JRC flood frequency, WorldPop population density, CHIRPS rainfall, SRTM elevation) to map cholera outbreak vulnerability at national scale.

**Eleyele Reservoir LULC - Optical vs SAR-Optical Fusion** *2026*

*Coursework assignment - Google Earth Engine, ArcGIS Pro*

- Compared Sentinel-2-only (97.09% OA) against Sentinel-2 + Sentinel-1 SAR fusion (95.60% OA) using Random Forest. Fusion improved water detection and resolved bareland/built-up confusion that optical bands alone missed.

**Spatiotemporal LULC Dynamics & Urban Growth Drivers, Edo State, Nigeria** *1999–2024*

*Coursework assignment - Google Earth Engine, ArcGIS Pro*

- Classified six Landsat epochs into five land cover classes; built Markov transition matrices and landscape metrics to explain conversion patterns. Built-up area grew 37% against 96% population growth.

**Eleyele Reservoir - Vegetation Encroachment & Water Surface Dynamics** *2026*

*Coursework assignment - ArcGIS Pro*

- Quantified a 59% decline in average open-water surface area (41.4 ha → 17.1 ha) and produced vegetation persistence and seasonal dynamics maps from nine years of time-series data.

## LEADERSHIP & VOLUNTEER EXPERIENCE

**Welfare Director** - Enactus FUTA *2024*

**Vice President** - URP Students' Association, FUTA *2022–2023*